

Evaluation of Clusters in DNA Methylation Array Data of Head Neck Squamous Cell Carcinoma

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Abstract

Head neck squamous cell carcinoma (HNSCC) is one of the most common malignant tumors. The identification of useful predictive or prognostic biomarkers is critical to more successful treatment in HNSCC. In this study, we obtained DNA methylation data of 528 HNSCC patients from Genomic Data Commons. By performing clustering on methylation data and examining the association between clustering and survival outcomes, we illustrated the considerable prognostic values from methylation for HNSCC. Using the clustering results and other clinical features, we further built a multivariate model for the survival outcome prediction.

All codes and reports generated from this study can be found on GitHub:

<https://github.com/yamachan18/stat6500>

1. Introduction and Objectives

1.1 Head Neck Squamous Cell Carcinoma

HNSCC is one of the most common malignant tumors, ranking sixth in incidence worldwide [1]. Like any other cancers, early diagnosis of HNSCC is one of the most crucial factors contributing to less extensive and more successful treatment and better prognosis. Therefore, identification of prognostic biomarkers is essential for HNSCC treatment.

1.2 Methylation

DNA methylation is a key epigenetic modification that is associated with gene expression regulation. Methylation of Cytosine-phosphate-Guanine (CpG) sites in promoter regions is a major molecular mechanism of gene silencing, which underlies cancer development and progression. In cancer study, testing for CpG methylation has emerged as a clinically relevant tool for tumor detection and prognostic prediction [2].

1.3 Objectives (Problem Statement)

In this study, we are interested in investigating the prognostic values of methylation for HNSCC by answering the following questions: **1) Can patients be grouped by methylation alone? 2) Are these clustering results associated with survival? 3) Can we build a multivariate model to predict the patients' survival outcomes?**

2. Data Analysis

2.1 Data Description: download and cleaning

Our dataset consists of 528 HNSCC patient samples with matched clinical annotations and methylation profiles from the Genomic Data Commons (GDC, <https://portal.gdc.cancer.gov/>). The basic patient demographic information can be found in Table 1.

Table 1. Patient demographic

Characteristic	N (%)	
Median age at diagnosis in year (range)	62(20-90)	
Gender	Male	386 (73%)
	Female	142 (27%)
Race	White	452 (86%)
	Black	48 (9%)
	Asian and others	13 (2.5%)
Vital status	Alive	358 (68%)
	Dead	170 (32%)
Tumor status	Tumor free	339 (64%)
	With tumor	139 (36%)

The methylation information is obtained from Illumina's Infinium HumanMethylation450 BeadChip, providing methylation information of more than 450,000 methylation sites. After cleaning and applying mean imputation to fill in missing values, a matrix of the top 1,000 most variably expressed CpGs (using standard deviation) were retained for further analysis. Because there were multiple entries in the clinical annotations table for one patient which needed to be formatted to have unique row names, it was wrangled such that we retained the information from the follow-up date that was furthest from the study start date.

2.2 Clustering Based on Methylation profiles

We applied clustering methods of k-means, partitioning around medoids (PAM), and spectral clustering to the methylation matrix that contains the top 1,000 most variably expressed CpGs. Specifically, we used the following clustering functions: *stats::kmeans*, *cluster::pam*, and *kernlab::specc*, and visualized the results of the clustering by coloring the t-distributed Stochastic Neighbor Embedding (TSNE) plot generated by the *Rtsne::Rtsne* function. The *clValid* package was used to compare internal validation metrics for k-means and PAM clustering for different numbers of clusters, k.

2.2.1 Challenges in choosing an optimum clustering scheme and number of clusters

When we fed in the original matrix of CpGs without transforming the beta values into *clValid*, we did not see an agreement between the internal cluster validation indexes. An example of an internal cluster validation index is the Silhouette width, which represents how well an observation fits into its assigned cluster and considers its distance to points in other clusters. Another internal validation index called the Dunn index, is the ratio of the minimum between-cluster distance to the maximum within-cluster distance [3]. For an ideal partitioning scheme, the average Silhouette width and Dunn index should be maximized. When evaluating PAM and k-means clustering with k ranging from 2 to 6, the largest average Silhouette width corresponded to k-means with k=2 (Fig. 1a), and the largest Dunn index corresponded to k-means with k=3 (Fig. 1b). Although the Silhouette widths do decrease between k=2 and k=3 for both k-means and PAM, this is a marginal drop, and the Dunn index increases quite drastically between k=2 and k=3. These patterns in the internal validation metrics hinted that k=3 may be the most optimum number of clusters.

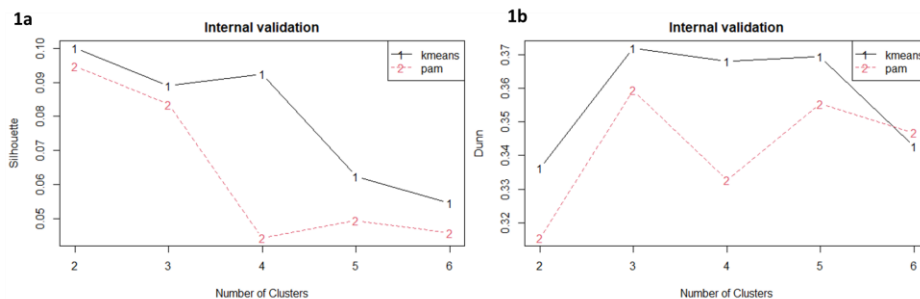


Figure 1. Plots showing relationship between internal validation metrics and number of clusters.

For spectral clustering, which performs well in identifying continuous shapes (e.g. spirals) and not necessarily data that is close together in the Euclidean space, a different heuristic, called the eigengap heuristic, is used to suggest the optimum value of k [4]. The eigengap heuristic states that k corresponds to the largest difference between eigenvalues of the Laplacian matrix, i.e. $\lambda_k - \lambda_{k+1}$. Unfortunately, the *kernlab* implementation of spectral clustering does not return the eigenvalues nor the Laplacian matrix from which the eigenvalues are computed, so we could not use the eigengap heuristic to inform us of the optimum number of clusters in spectral clustering.

Judging from the results of *clValid*, we expected to see a 3-part structure of our data. Visualizing our high-dimensional matrix in a two-dimensional embedding using TSNE allowed us to see a rough structure of our data, and that it could possibly be partitioned into 4, possibly 5 different groups (Fig. 2a), which contrasted with what we expected from the result of *clValid*. However, none of the algorithms were able to identify this potential 4- or 5-part structure (Fig. 2b-f).

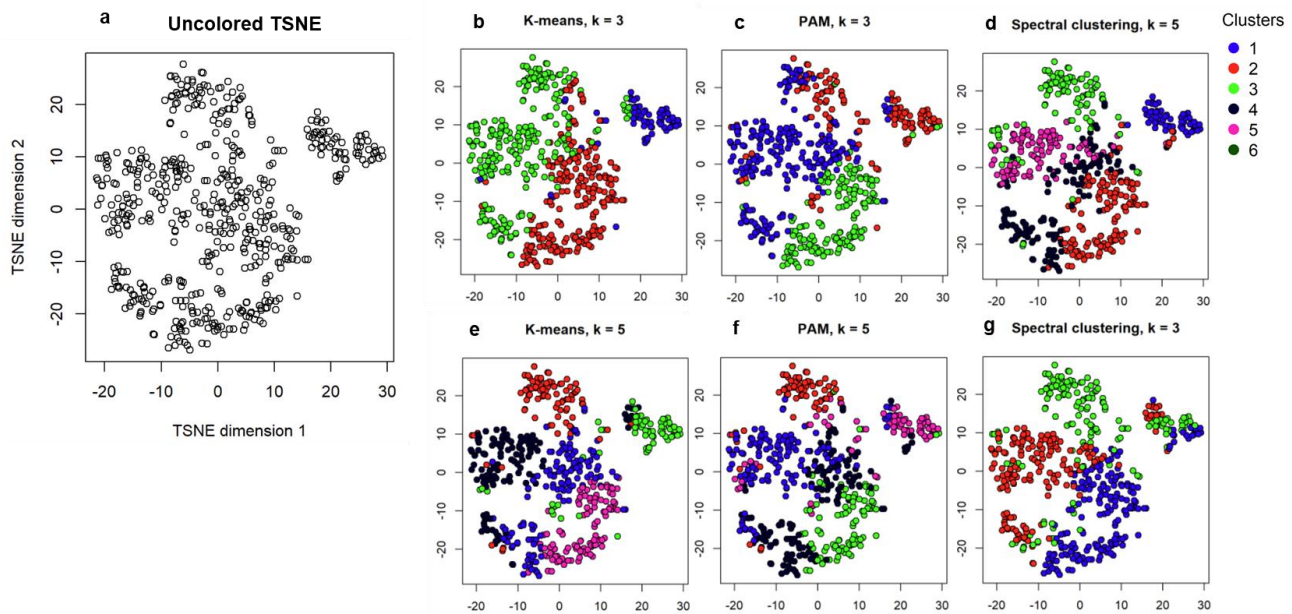


Figure 2. Low-dimensional embedding of methylation array data colored by cluster calls

We also attempted simulating clusters by drawing from a random beta distribution, assigning the rows to a cluster, and shifting the values by some arbitrary constant. Since we knew what the correct clustering scheme should be,

in theory, we would be able to choose a clustering scheme based on what was closest to our simulated cluster calls. However, the process of simulating our clusters, namely the shifting of the values by a constant, made it such that k-means outperformed every other clustering scheme. Therefore, simulation was abandoned for our goal of choosing a clustering scheme and an appropriate k value. In summary, the clustering algorithms we employed on the untransformed data were also not able to identify the structure we saw from plotting the low-dimensional representation of the data, which made us hesitant about choosing a specific clustering result over another.

2.2.2 Alternative approach: using the Jaccard distance

We tried an alternative approach, in which we converted our matrix of CpGs into a Jaccard distance matrix using the *binaryDistance* function from the *Mercator* package. The Jaccard distance, which is usually computed for binarized values, is a distance metric that measures the dissimilarity between sample sets [5]. Although our data is not completely binary, is distributed between 0 and 1 and skewed, such that it is still mathematically reasonable to apply the Jaccard distance. After computing the Jaccard distances, we used this $n \times n$ matrix for TSNE and clustering. From visual inspection of the uncolored TSNE (Fig. 3a) we observed a three-part structure, and this structure could only be identified by PAM with $k=3$ (Fig. 3b).

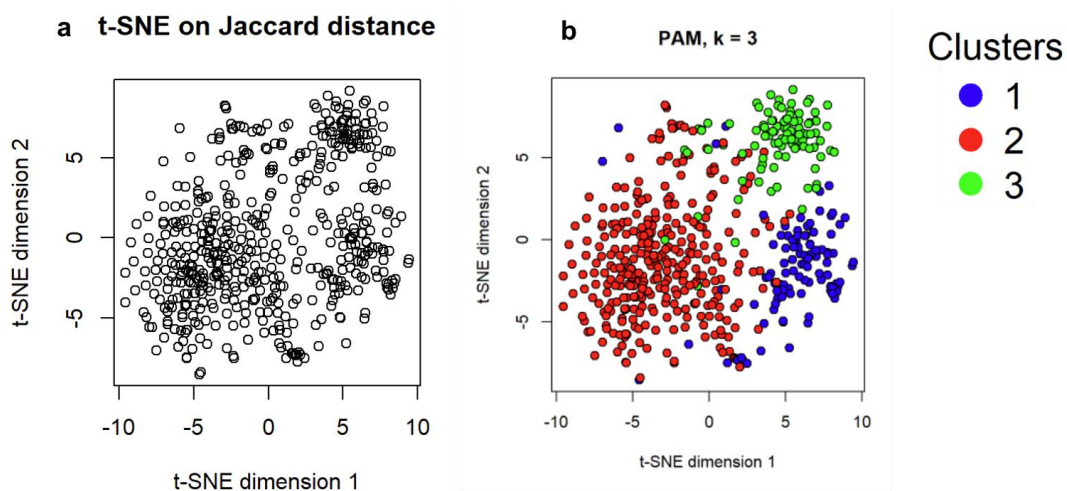


Figure 3. Low-dimensional embedding of the Jaccard dissimilarity matrix of methylation array data colored by cluster calls

2.3 Survival Analysis

To investigate the factors that influence overall survival (OS), we performed survival analysis based on the clustering results as well as other clinical features. The R package of *survival* and *survminer* were applied. Specifically, *surfit* function computes an estimate of a survival curve for censored data using the Kaplan-Meier (KM) method. *Coxph* fits a Cox proportional hazards regression model and calculates the hazard ratio (HR). HR below 1 indicates a covariate that is negatively associated with the event probability, and thus positively associated with the length of survival.

2.3.1 Survival Analysis Using Methylation Clustering

We conducted survival analysis on the all clustering analysis. Among all clustering methods, PAM with k value of 3 (referred as PAM3) when using the Jaccard distance for TSNE on methylation data shows clear association with OS (Fig 4). Specifically, the patients with methylation profile in cluster 3 have the best survival outcomes, as suggested by higher survival probability and HR value (below 1). On the contrary, the patients with methylation profile in cluster 1 tend to have worse survival outcomes.

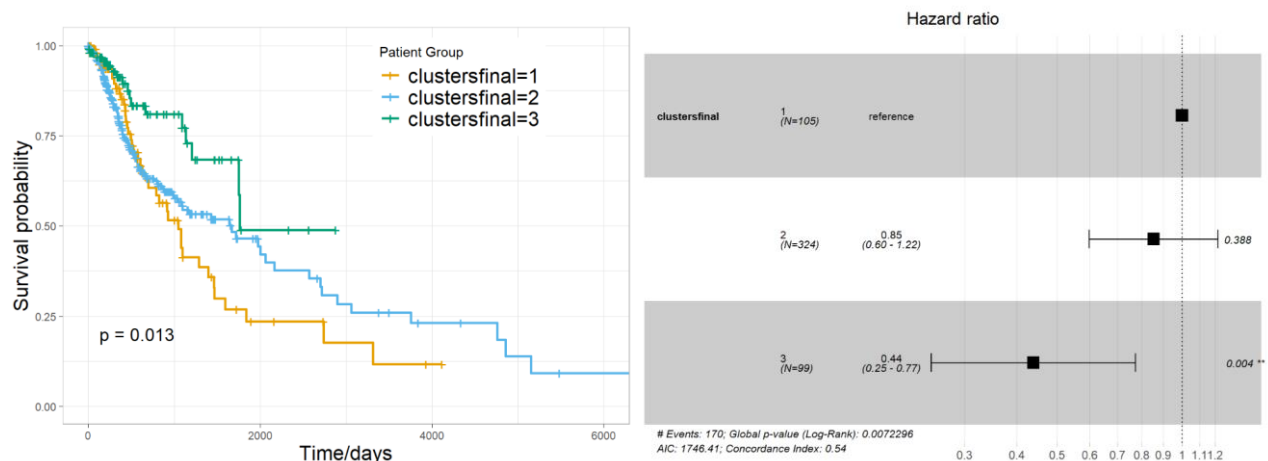


Figure 4. KM survival curve and forest plot for survival analysis using methylation clustering (PAM3)

2.3.2 Survival Analysis Using Single Clinical Features

We checked the association between the survival and clinical variables containing less than 6 unique levels, including gender, age groups, race, ethnicity, tumor status, tumor grade, history of other malignancy, AJCC staging

information, etc. Among these features, gender, age groups, tumor status and tumor grade show significant association with survival outcomes (Fig). Notably, the performance of methylation clustering is even better than some clinical features such as age group and tumor grade, as demonstrated by the smaller p value, which further illustrates the prognostic value of methylation data in HNSCC.

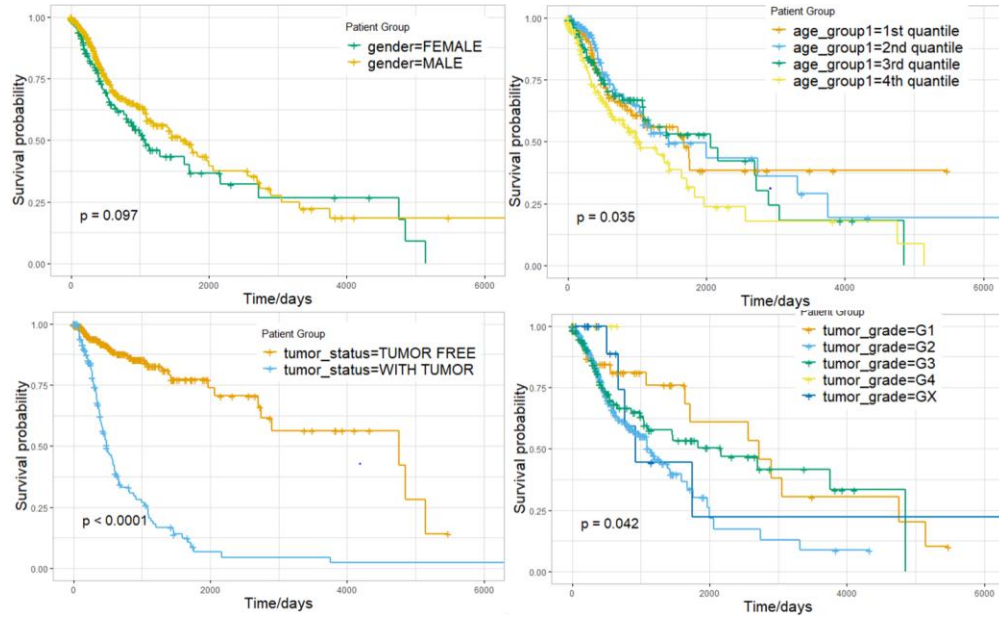


Figure 5. KM survival curve for clinical features of gender, age group, tumor status and tumor grade.

2.4 Multivariate Predictive Modeling

We are interested in building a model to predict the patient survival outcomes (alive as event 0, dead as event 1). The patient samples were randomly split into training and test (50% and 50%) set. Then a simple multivariate predictive model using linear discriminant analysis (LDA) method was constructed on training set, with input of methylation-based clustering results (PAM3) together with clinical variables of gender, age group, tumor status and tumor grade. This model was tested on the test data set and achieved a test error of 0.2. The Fig 6 shows the predictive survival outcomes on test set, comparing to real survival outcomes annotated in different colors.

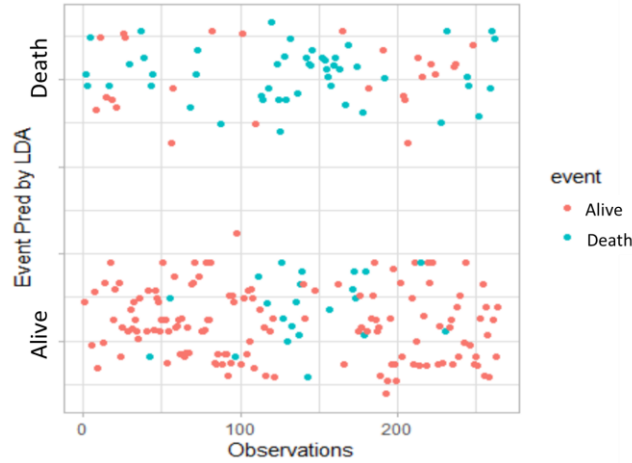


Figure 6. Predictive performance of the LDA model

3. Conclusion and Discussion

In this study, we firstly constructed several subgrouping/clustering models (K-means, PAM and spectral clustering) using only the methylation data of the top 1,000 most variably expressed CpGs, and then assessed the clustering performance. Additionally, we investigated the association between methylation-based patient clusters and survival outcomes. Survival analysis using other clinical features such as age group and tumor grades was also included and compared. Finally, we built a multivariate model to predict the HNSCC patient survival outcomes using methylation-based patient clustering and a few clinical features.

During the clustering analysis, we identified three putative clusters by PAM with K value of 3 (referred as PAM3), when using the Jaccard distance for TSNE on methylation data. Notably, these three clusters show strong association with overall survival (OS), as suggested by small p value (0.013) and HR value between different clustering groups. Moreover, compared to some clinical variables, the association between PAM3 clustering and survival is even more notable. In practice, the HNSCC patients with methylation profiles similar to the first PAM3 clustering could be stratified at diagnosis as high-risk patients who need more aggressive treatment.

Lastly, we constructed survival outcome predictive model using LDA method with input of methylation-based clustering results (PAM3) together with clinical variables of gender, age group, tumor status and tumor grade. The

test error is around 0.2 when predicting the survival outcomes (dead or alive). The performance of the predictive modeling can be further improved. For example, we may try other variable inputs and then conduct stepwise variable selection. Assigning different weights to the selected predictors might also help to improve the modeling performance.

In summary, we have identified three methylation-based clusters to group the HNSCC patients. This clustering result is closely associated with survival outcome and is useful in building a multivariate model to predict the patient survival outcomes. This study signifies the prognostic value of methylation data in HNSCC study.

References

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